

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Civil Engineering
2. **Subject Name:** Probability statistics and inference in civil engineering **L-T-P:** 3-0-0
Credits: 3
3. **Pre-requisites:** None
4. **Syllabus and reference books:**

Syllabus: Motivation, problems in Civil Engineering & basic definitions of probability, random variables and distributions; role of uncertainty in various disciplines of Civil Engineering; practical examples in structural, geotechnical, transportation, hydrological and environmental engineering with data, types of uncertainty in Civil Engineering, joint distributions and joint hazard modeling; Common discrete and continuous distributions in Civil Engineering and how they arise (e.g., in return period of earthquakes and floods, runoff, material strength, dead and live loads, soil capacity, traffic flow, concentration of pollutants etc.); Monte Carlo simulations of univariate (discrete and continuous) and dependent random variates for estimating member strengths, time-dependent loads, rare probabilities etc.; Descriptive statistics, estimation, hypothesis set up and inference, model selection and regression in structural, geotechnical, transportation, environmental, hydroclimatologic etc. problems; Application in various topics in Civil Engineering namely random vibration, natural hazard modeling, public health engineering, structural reliability, structural health monitoring and system identification.

Reference Books / Articles:

Required:

- Probability Concepts in Engineering: Emphasis on Applications to Civil and Environmental Engineering, A HS Ang and W H Tang, Wiley 2006

Recommended:

- *Probability Statistics and Decision for Civil Engineers*, Benjamin and Cornell, McGraw Hill, 1970.
- *Uncertainty Modeling with Applications to Multidimensional Civil Engineering Systems*, O. Ditlevsen, McGraw Hill, 1981.
- *Probability and Statistics Applications for Environmental Science*, Shaefer and Theodore, CRC Press, 2007.
- *Probability Random Variables and Stochastic Processes*, 4/ed., Papoulis and Pillai, □ McGraw Hill, 2017.
- *Structural Reliability Analysis and Prediction*, 3/ed, Melchers and Beck, Wiley, 2018.
- *Statistical Methods in Hydrology and Hydroclimatology*, R. Maity, Springer, 2018

5. **Lecture-wise break-up:**

1	Role of uncertainty in various disciplines of Civil Engineering; practical examples in structural, geotechnical, transportation, hydrological and environmental engineering with data, definitions and axioms of probability, probability of joint events, types of uncertainty in Civil Engineering, random variables and properties of distribution functions, joint distributions for joint hazard modeling	6
2	How various discrete and continuous distributions arise and their relevance to various topics in Civil Engineering, e.g., return period of earthquakes and floods, runoff, material strength, dead live and extreme loads, soil capacity, traffic flow, turbulence, concentration of pollutants, population forecasting etc.	6
3	Monte Carlo simulations: pseudo random number generators, simulation of binary events and their aggregates, inverse CDF method, simulation of univariate and dependent variates for traffic flow, hydroclimatologic phenomena, strength of steel and reinforced concrete sections, time-dependent load models, estimation of rare probabilities, etc.	6
4	Summarizing and displaying data sets, histograms, cumulative frequency, point estimation, interval estimation, Hypothesis set up and one side and two sided tests, Type I and Type II errors in structural, geotechnical, transportation, environmental, hydroclimatologic etc. problems	6
5	Model selection and linear, polynomial and multivariate regression in structural, geotechnical, transportation, environmental, hydroclimatologic etc. problems	6
6	Applications in Civil Engineering selected from (with emphasis on): Random vibrations (stationary stochastic processes), Natural hazard modeling (asymptotic theory of extremes), Water resources (risk management), Public health engineering (analysis of variance), Structural reliability (factors of safety and limit states), Structural health monitoring and system identification (Bayesian updating and linear filtering)	6
	Total	36